

① Ejercicios :

a) Partícula libre ✓

b) operadores ✓

c) Átomo de Hidrogeno

Partícula libre ✓

$e^- \xrightarrow{\vec{v}_0}$ $\vec{F} = \vec{0}; \quad \vec{F} = -\vec{\nabla} E_p \quad E_p = 0$

Ecuación de Schrödinger :

$$\hat{H} \phi = E \phi; \quad \hat{H} = \hat{E}_K + \hat{E}_p$$

$$\textcircled{2} \quad \hat{H} = \frac{p^2}{2m_e} + E_p$$

$$E_p = 0;$$

$$\hat{H} = - \frac{\hbar^2 \nabla^2}{2m_e};$$

$$- \frac{\hbar^2 \nabla^2}{2m_e} \phi = E \phi$$

$$\textcircled{3} -\frac{\hbar^2 \nabla^2 \phi}{2me} = E \phi$$

$$\textcircled{1} \nabla^2 \phi + k^2 \phi = 0; \quad k^2 = \frac{2meE}{\hbar^2}$$

Separación de Variables:

$$\phi(x, y, z) = \chi(x) \gamma(y) \zeta(z) \quad \textcircled{2}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (\underbrace{\chi(x) \gamma(y) \zeta(z)}) + k^2 \chi \gamma \zeta = 0$$

$$\gamma(y) \zeta(z) \frac{\partial^2 \chi(x)}{\partial x^2} + \chi(x) \zeta(z) \frac{\partial^2 \gamma(y)}{\partial y^2} + \chi(x) \gamma(y) \frac{\partial^2 \zeta(z)}{\partial z^2} + k^2 \chi \gamma \zeta = 0$$

$$\frac{\cancel{\gamma(y)} \cancel{\zeta(z)}}{\chi \cancel{\gamma} \cancel{\zeta}} \frac{d^2 \chi(x)}{dx^2} + \frac{\cancel{\chi(x)} \cancel{\zeta(z)}}{\cancel{\chi} \gamma \cancel{\zeta}} \frac{d^2 \gamma}{dy^2} + \frac{\cancel{\chi(x)} \gamma(y)}{\cancel{\chi} \cancel{\gamma} \zeta} \frac{d^2 \zeta}{dz^2} + k^2 \frac{\cancel{\chi} \cancel{\gamma} \cancel{\zeta}}{\cancel{\chi} \cancel{\gamma} \cancel{\zeta}} = 0$$

$$\underbrace{\frac{1}{\chi(x)} \frac{d^2 \chi(x)}{dx^2}}_{-k_x^2} + \underbrace{\frac{1}{\gamma(y)} \frac{d^2 \gamma}{dy^2}}_{-k_y^2} + \underbrace{\frac{1}{\zeta} \frac{d^2 \zeta}{dz^2}}_{-k_z^2} + k^2 = 0$$

$$\frac{1}{\chi(x)} \frac{d^2 \chi(x)}{dx^2} = -k_x^2 \quad \textcircled{3}$$

$$\frac{1}{\gamma(y)} \frac{d^2 \gamma(y)}{dy^2} = -k_y^2 \quad \textcircled{4}$$

$$\frac{1}{\zeta(z)} \frac{d^2 \zeta(z)}{dz^2} = -k_z^2 \quad \textcircled{5}$$

$$k^2 = k_x^2 + k_y^2 + k_z^2$$

$$\textcircled{+} \frac{1}{\chi(x)} \frac{d^2 \chi(x)}{dx^2} = -k_x^2$$

$$\frac{d^2 \chi(x)}{dx^2} + k_x^2 \chi(x) = 0$$

$$\chi(x) = A e^{i k_x x} + B e^{-i k_x x}$$

$i = \sqrt{-1}$

$$Y(y) = C e^{i k_y y} + D e^{-i k_y y}$$

$$Z(z) = E e^{i k_z z} + F e^{-i k_z z}$$

$$\phi(x, y, z) = (A e^{i k_x x} + B e^{-i k_x x}) (C e^{i k_y y} + D e^{-i k_y y}) (E e^{i k_z z} + F e^{-i k_z z})$$

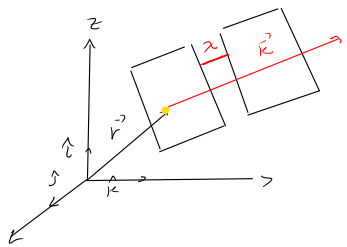
$$\psi(x, y, z, t) = e^{-i \frac{E}{\hbar} t} \phi(x, y, z)$$

$$\psi(x, y, z, t) = (A e^{i(k_x x - \omega t)} + B e^{-i(k_x x - \omega t)}) \dots$$

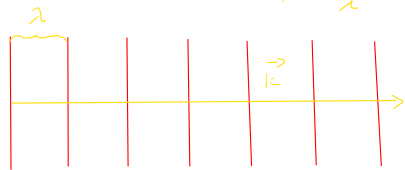
$$\psi(x, y, z, t) = R e^{i(\vec{k} \cdot \vec{r} - \omega t)} + M e^{-i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\vec{k} = k_x \hat{x} + k_y \hat{y} + k_z \hat{z}$$

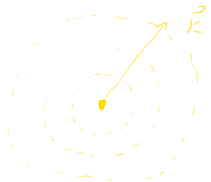
$$\vec{r} = x \hat{x} + y \hat{y} + z \hat{z}$$



$$|\vec{k}| = \frac{2\pi}{\lambda}$$

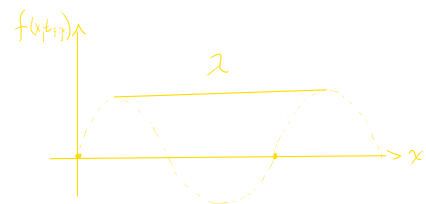


$$|\vec{k}| = k_x^2 + k_y^2 + k_z^2$$



$$1D: f(x, t) = A e^{i(kx - \omega t)} + B e^{-i(kx - \omega t)}$$

$$f(x, t) = A \sin(kx - \omega t)$$



② operadores diferenciales
Ejercicio: ¿cuáles de las siguientes funciones
son autofunciones del operador momento lineal?

- $\vec{p}$
- a) $A \sin(kx)$, b) $A \cos(kx) + i A \sin(kx)$
c) $A \sin(kx) - A \cos(kx)$ d) $A e^{ik(x-a)}$

Solución:

$$\hat{p} = -i\hbar \vec{\nabla}$$

$$\hat{p} = \hat{p}_x \hat{e} + \hat{p}_y \hat{j} + \hat{p}_z \hat{k}$$

$$\hat{p}_x \hat{e} + \hat{p}_y \hat{j} + \hat{p}_z \hat{k} = -i\hbar \frac{\partial}{\partial x} \hat{e} - i\hbar \frac{\partial}{\partial y} \hat{j} - i\hbar \frac{\partial}{\partial z} \hat{k}$$

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}; \hat{p}_y = -i\hbar \frac{\partial}{\partial y}; \hat{p}_z = -i\hbar \frac{\partial}{\partial z}$$

$$f(x) = A \sin(kx)$$

$$\begin{aligned} p_x f(x) &= -i\hbar \frac{\partial}{\partial x} (A \sin(kx)) \\ &= -i\hbar \frac{d}{dx} (A \sin(kx)) \\ &= -i\hbar k A \cos(kx) \end{aligned}$$

$$p_x (A \sin(kx)) = -i\hbar k A \cos(kx)$$

$$f(x) = A \sin(kx) \text{ no es autofunción}$$

$$p_x \phi = \lambda \phi; \quad \begin{cases} \phi: \text{Autofunción} \\ \lambda: \text{Autovalor} \end{cases}$$

$$f_2(x) = A \cos(kx) + i A \sin(kx)$$

$$\begin{aligned} \hat{p}_x f_2(x) &= -i\hbar \frac{\partial}{\partial x} (A \cos(kx) + i A \sin(kx)) \\ &= +i\hbar A k \sin(kx) + \hbar A k \cos(kx) \\ &= \hbar k (A \cos(kx) + i A \sin(kx)) \\ &= \hbar k f_2 \end{aligned}$$

f_2 : Autofunción del operador momento

$$\text{Autovalor } \lambda = \hbar k \quad f_2 = A e^{ik(x-a)}$$

$$f_3 = A \sin(kx) - A \cos(kx)$$

$$\begin{aligned} \hat{p} f_3 &= -i\hbar \frac{\partial}{\partial x} (A \sin(kx) - A \cos(kx)) \\ &= -i\hbar k A \cos(kx) - i\hbar k A \sin(kx) \\ &= -i\hbar k (A \cos(kx) + A \sin(kx)) \end{aligned}$$

f_3 : no es autofunción

$$f_4 = A e^{ik(x-a)}$$

$$\begin{aligned} \hat{p}_x f_4 &= -i\hbar \frac{\partial}{\partial x} (A e^{ik(x-a)}) \\ &= ik A e^{ik(x-a)} (-i\hbar) \\ &= \hbar k A e^{ik(x-a)} \\ &= \hbar k f_4 \end{aligned}$$

⑥ Ejercicio: Un electrón se describe en la función de onda

$$\phi(x) = \begin{cases} 0 & \text{para } x < 0 \\ c e^{-x} (1 - e^{-x}) & x \geq 0 \end{cases}$$

donde x está en nanómetros y C es una constante.

a) Encontrar la constante de normalización C

b) Realizar una gráfica de la función de probabilidad ✓

$$p(x) = |\phi|^2$$

c) ¿Dónde es más probable encontrar al electrón? es decir, para qué valor de x es máxima la probabilidad

Solución

$$\int_{-\infty}^{\infty} \phi \phi^* dx = 1$$

$$\int_{-\infty}^0 \cancel{\phi \phi^*} dx + \int_0^{\infty} \phi \phi^* dx = 1$$

$$\int_0^{\infty} \phi \phi^* dx = 1$$

$$\int_0^{\infty} (c e^{-x} (1 - e^{-x})) (c^* e^{-x} (1 - e^{-x})) dx = 1$$

$$|c|^2 \int_0^{\infty} e^{-x} (1 - e^{-x}) e^{-x} (1 - e^{-x}) dx = 1$$

1

$$|c|^2 = \frac{1}{\int_0^{\infty} e^{-x} (1 - e^{-x}) e^{-x} (1 - e^{-x}) dx}$$